Damien LE QUÉRÉ
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website – linkedin



R&D Engineer
Software and Telecom

During last years I worked on Real-Time telecom systems. I am currently looking for a job in the continuity of my last experiences. I am specially interested to use and improve my skills in C++ real-time development.

Software engineering

- Real-time development: **C++99**
- Few experiences : C++14, boost

Network Architectures

- Mobile Networks: **LTE (4G)**
- Transport Networks : MPLS, PBB-TE

Tools

- Edition: **VS Code**, **Emacs**, CLion
- Building: CMake
- VCS / Review: **Git** / **Gerrit**
- CI / Delivery: **Jenkins** / Zuul

- Design : UML (PlantUML)
- System : Debian

Experience

R&D Software engineer (4G)

2018 – 2020 (2.5 years)
CDI in Lannion (22)

Work on a *Nokia 4G multiple UEs simulator*. Embedded development and maintenance of L2 processes launched on DSPs: UE Scheduler, DL HARQ, DL MAC. Performance investigations to reach maximum throughputs. Investigations about various U-plan issues. Agile software development: SCRUM.
Contribution to reach 2.1Gbps in downlink on 4UEs (massive MIMO with 20 layers).
Contribution to ARM porting (Texas Instrument: Multicore DSP+ARM KeyStone II System-on-Chip).

– embedded development: C++99 (on Texas Instrument DSPs)
– version control / code review: Git / Gerrit
– continuous integration / delivery: Jenkins / Zuul
– automatic testing / scripting : lua + robot + bash

Telecom system integrator

2016 – 2017 (1.5 year)
CDI in Nozay-Villarceaux (91)

Intégration of *Nokia 9500 Microwave Packet Radio*. Integration of the modem board, radio board, and assembled product (OutDoor Unit). Method: SCRUM.

– network management : Nagvis/Nagios, Opengear console server, scripting, telnet, SSH
– telecom / embedded software : Broadcom (vxWorks), Eband (80GHz), FPGA (Field-Programmable Gate Array), GPIO (General Purpose Input/Output), Spectrum Analyser (Anritsu)
– automatiques tests : C++ (CppUnit), Python
– continuous integration : Jenkins, JIRA

Telecom researcher (PhD student)

2011 – 2015 (3 years)
Thesis in Lannion (22)

Industrial thesis at *Orange Labs* supervised by Adlen Ksentini / Yassine Hadjadj-Aoul (IRISA (Institut de Recherche en Informatique et Systèmes Aléatoires)) and Christophe Betoule (Orange Labs). Subject : "*Design and performances study of a self-configurable solution for future transport networks*".

Study of the packet architecture *LOCARN* (Low Opex Capex Architecture for Resilient Networks) in the operator transport ecosystem (IP, MPLS, Carrier Ethernet, OTN / WDM). Positioning of the solution compared to the *GMPLS* control plane, *Autonomic Networking* and *SDN*. Complete development of a simulation model in *C++* and *Python* (OMNeT ++ network simulator). Contribution with two architectural proposals improving the scalability performances of the solution.

Telecom intern

2011 – 2015 (6 months)
Internship in Lannion (22)

Internship within the *Envision* project. Development of a stream translation system allowing a network operator to benefit from multicast for punctual events, transparently or in collaboration with an overlay application. Complete demonstrator implementation with the GNU/Linux kernel, initiation to the Junos-SDK framework (C / C ++). Definition of test cases, traffic simulation with N2X servers, design and development of the multicast controller (Java) and redaction of the documentation.

– networking: NAT, filtering, static multicast routing
– protocols: PIM SM SSM, IGMPv3, ISIS
– tools: N2X Server (traffic simulation), *iptables*, *smcroute*, SSH, *Juniper M7i* Router, *Junos SDK 10.4*

Software intern

2010 (6 months)
Internship in Chinon (37)

Intership in the *Endel Pôle Nucléaire* group
Development / deployment of *Aguenau*, a site management application. Software maintenance and extensions according to users feedbacks. Microsoft technologies: *Terminal Service*, *VBA* and *Transact-SQL*, study of the network architecture and of the potential deployment at the national level. Result: operational application for the two mission's sites: Chinon and Chooz.

Education

PhD in Computer Science

2012 – 2015

« Diplôme de docteur de l'université Rennes 1, mention informatique, mention très honorable »
Degree from the Rennes 1 University, CIFRE (Convention Industrielle de Formation par la REcherche) grant at Orange Labs (first class honours)

Master degree in CS (Networking)

2010 – 2011

« Master 2 pro informatique : Ingénierie des Réseaux »
Degree from the *Rennes 1 University* (with distinctions)

Master degree in CS (Software Engineering)

2009 – 2010

« Master 2 pro informatique : Conception et Développement de Solutions Informatiques Intégrées »
Degree from the *Angers University of Sciences* (with distinctions)

Master in CS (Software Architecture)

2008 – 2009

« Master 1 pro/recherche informatique : ALMA option Architecture Logicielle »
Year validated at the *Nantes University of Sciences and Techniques*

Publications

International Workshop

D. Le Quere, C. Betoule, R. Clavier, Y. Hadjadj-Aoul, A. Ksentini, and G. Thouenon.
« Towards a Large Scale LOCARN ».
IEEE Globecom Workshop on Research and Standards (Dec 2014, Austin, United States).

Journal (Special Issue)

D. Le Quere, C. Betoule, R. Clavier, Y. Hadjadj-Aoul, A. Ksentini, and G. Thouenon.
« Presentation & evaluation of LOCARN: low opex and capex architecture for resilient networks ».
Studia Informatica Universalis, pages 74-104 (12 June 2014).

International Conference

Best Paper Award
D. Le Quere, C. Betoule, R. Clavier, Y. Hadjadj-Aoul, A Ksentini, and G. Thouenon.
« Scalability & performances evaluation of LOCARN: low opex and capex architecture for resilient networks ».
Innovations for Community Services (I4CS 2014), 14th International Conference, pages 1–8 (June 2014, Reims, France).

Languages

- English : CLES (Certificat de Compétences en Langues de l'Enseignement Supérieur) level 2 (TOEIC (Test Of English for International Communication) equivalent score = 785 points)

Interests

- Web development:
  - Stack: *React*, *TypeScript*, *GraphQL*, *URQL/Apollo*, *Node.js*, *PostegreSQL*, *TypeORM*, *Redis*, *Next.js*
  - Other: *Semantic web*, *SVG*, *Canvas*
- *Emacs* / *Lisp*
- Edition: LaTeX, *Gnuplot*, *Inkscape*
- Sport : cycling, windsurfing, badminton (7 years)
- Music : piano beginner, guitar (7 years)